

Business Summary Report

Customer Delinquency Prediction – Geldium
Tata GenAI Powered Data Analytics Job Simulation – Task 3

1. Summary of Predictive Insights

The Logistic Regression model (86% accuracy) built in Task 2 highlights the following patterns among Geldium's customers:

- **Payment behaviour is the strongest signal.** Customers with a higher number of Missed_Payments in the past 12 months, and a pattern of "Late" or "Missed" statuses across Month_1–Month_6, are far more likely to be flagged as delinquent.
- **Financial stress indicators compound risk.** High Credit_Utilization combined with a high Debt_to_Income_Ratio and a low Credit_Score identifies a clearly higher-risk segment than any single factor alone.
- **Roughly 1 in 6 customers is high-risk.** About 16% of customers in the dataset are delinquent, so the Collections team should expect a meaningful but manageable share of the portfolio to need proactive attention rather than reactive follow-up.

Key Insights Summary

Key Insight	Customer Segment	Influencing Variables	Potential Impact
Customers with 4+ missed payments and rising credit utilization are at sharply higher delinquency risk.	High missed-payment, high-utilization customers	Missed_Payments, Credit_Utilization, Month_1–Month_6 status	Early flagging enables outreach before the account becomes delinquent
Low credit score combined with high debt-to-income ratio strongly predicts delinquency.	Financially stretched, lower credit-score customers	Credit_Score, Debt_to_Income_Ratio, Loan_Balance	Supports risk-based credit limit and repayment-plan decisions
Recent payment trend (last 2–3 months) is more predictive than older history.	Customers showing recent "Late"/"Missed" status	Month_4, Month_5, Month_6	Enables a real-time early-warning trigger for Collections

2. Recommendation Framework

Restated Insight: Customers with 4 or more missed payments in the past 12 months, combined with rising credit utilization and a recent (last 2-3 month) pattern of late or missed payments, are significantly more likely to become delinquent.

Proposed Recommendation: Launch a proactive early-outreach programme that automatically flags and contacts customers matching this risk profile before they miss another payment.

Specific: Target customers with Missed_Payments ≥ 4 AND at least one "Late" or "Missed" status in Month_5 or Month_6, offering a payment reminder call plus an optional revised repayment plan.

Measurable: Track the delinquency rate of the flagged segment before vs. after the programme, and aim to reduce new delinquencies within this segment by at least 15% over one quarter.

Actionable: The model's monthly risk score can be pushed directly into the Collections team's existing outreach queue/CRM, so no new infrastructure is required to act on it.

Relevant: Directly addresses the strongest predictors identified by the model (missed payments and utilization trend), focusing effort on the segment most likely to benefit.

Time-bound: Pilot the programme for one quarter (3 months) on a sample of flagged customers, then review results and decide on full rollout.

Justification and Business Rationale: Reaching customers before they miss a payment is considerably cheaper and less damaging to the customer relationship than post-delinquency collections. Since payment-history and utilization variables are the strongest predictors in the model, targeting outreach specifically at this segment concentrates limited Collections resources on the customers where intervention is most likely to prevent an actual delinquency, rather than contacting the portfolio broadly.

3. Ethical and Responsible AI Considerations

- **Potential for bias:** Because Employment_Status and Location are included as predictors, the model could indirectly disadvantage customers in lower-income regions or non-traditional employment (e.g., self-employed or unemployed customers), even if that was not intended. For example, an "Unemployed" customer with an otherwise healthy repayment history could be scored as higher-risk purely due to employment status. Subgroup performance should be reviewed regularly to check for this.
- **Explainability:** Logistic Regression's coefficients make it possible to clearly explain to a customer or regulator which specific factors (e.g., missed payments, utilization) drove a given risk score, which supports transparent, defensible decision-making.
- **Responsible financial decision-making:** The recommendation is designed to trigger supportive outreach (reminders, flexible repayment options) rather than punitive action such as automatic credit restriction, so the model is used to help customers stay on track rather than penalize them pre-emptively.
- **Other principles:** Customer financial data should be accessed only by authorised staff and used solely for risk assessment (data privacy); model outputs should remain a decision-support input reviewed by a human agent rather than a fully automated decision (accountability); and model performance and fairness should be re-evaluated periodically as customer behaviour evolves (ongoing transparency).